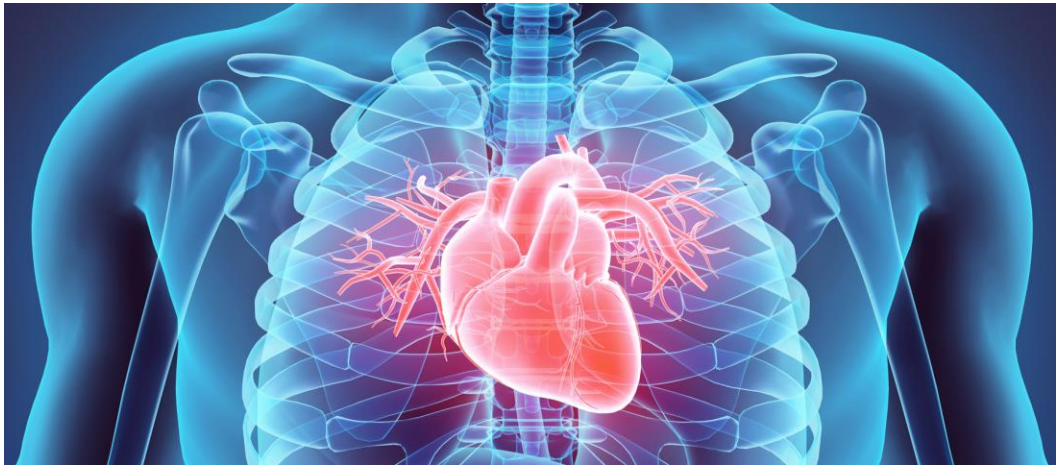
Heart, Lung and Critical Care Clinical Group

NHS
Guy's and St Thomas'
NHS Foundation Trust

Royal Brompton Hospital
Harefield Hospital

Cardiovascular Anatomy
Heart & Mediastinum



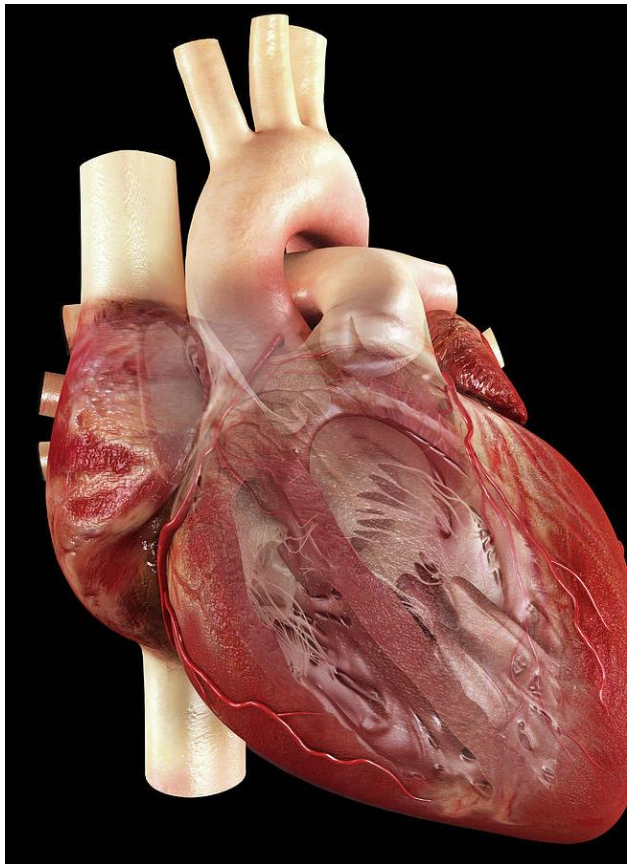
Beccy Hall

February 2025

LONDON METROPOLITAN UNIVERSITY

Learning Outcomes:

- Be able to identify key surface anatomy of the heart giving a brief account of the orientation of this organ
- Describe the features of the heart chambers
- Know the structure and function of the heart valves and know their positions for auscultation
- Describe the layers of the heart wall and pericardium
- Name the coronary arteries and their branches
- Define the mediastinum and list the content in its divisions and nerve supply



The cardiovascular system

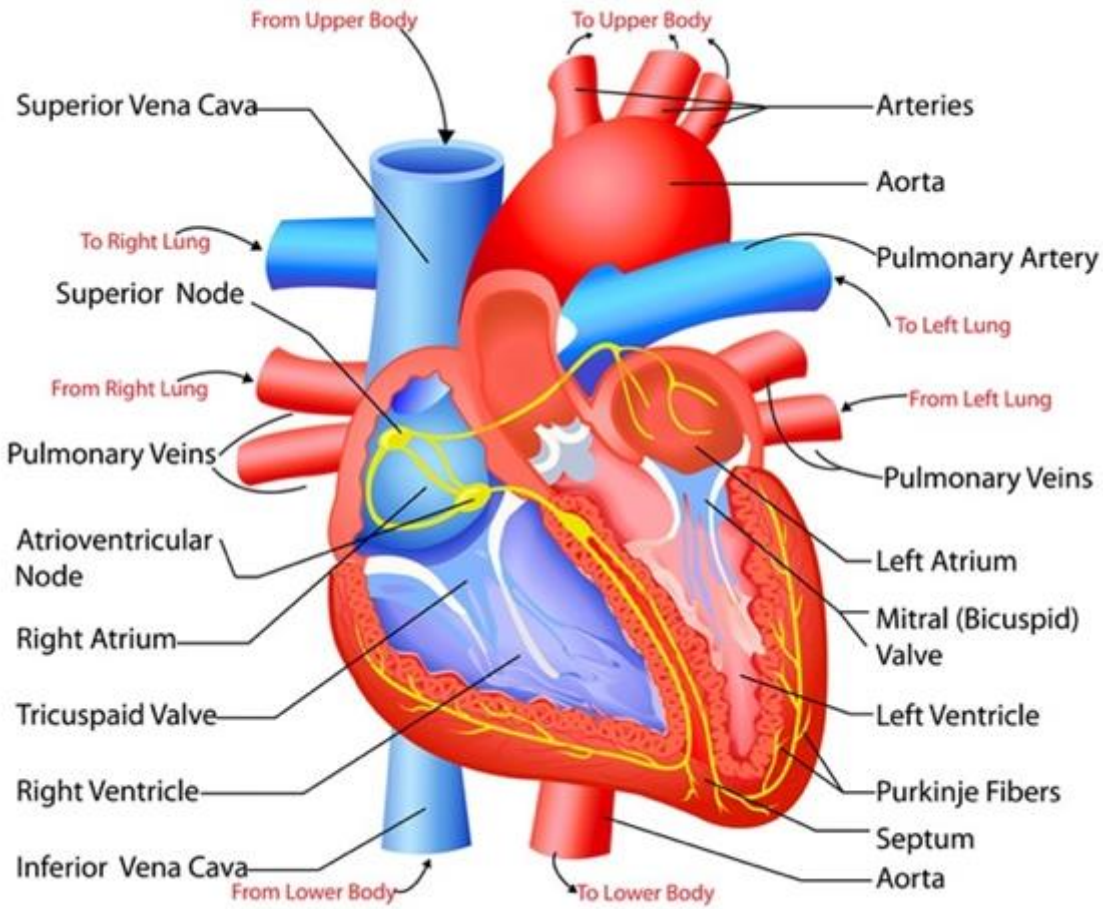
Circulatory system is made up of 3 components:

- **Heart** – establishes a pressure gradient by putting pressure on the blood, allowing it to flow to the tissues
- **Blood vessels** – Passageways through which blood is distributed around the body
- **Blood** – Transport medium – which the vital materials are suspended in. Including: O2, CO2, nutrients, electrolytes, hormones and waste products



Components of the Cardiovascular System

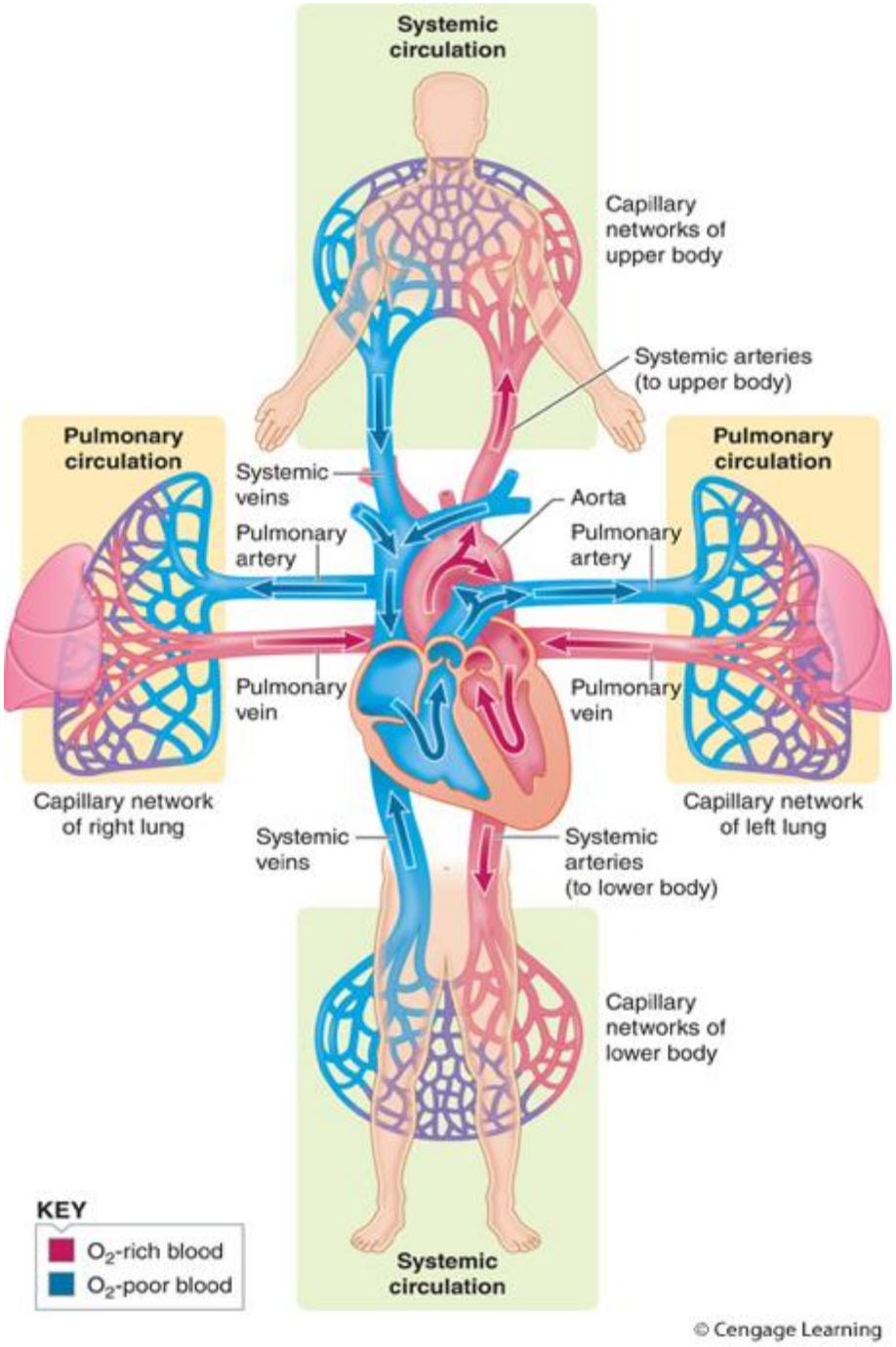
- Heart
- Blood Vessels
- Blood



Summary of Circulation:

2 separate vascular loops:

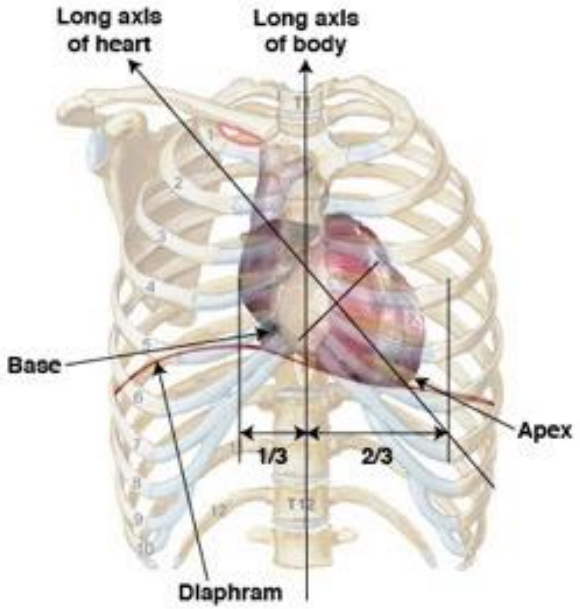
- **Pulmonary Circulation**
  - Closed loop
  - Carries blood between heart & lungs
- **Systemic Circulation**
  - Carries blood between heart and all body systems

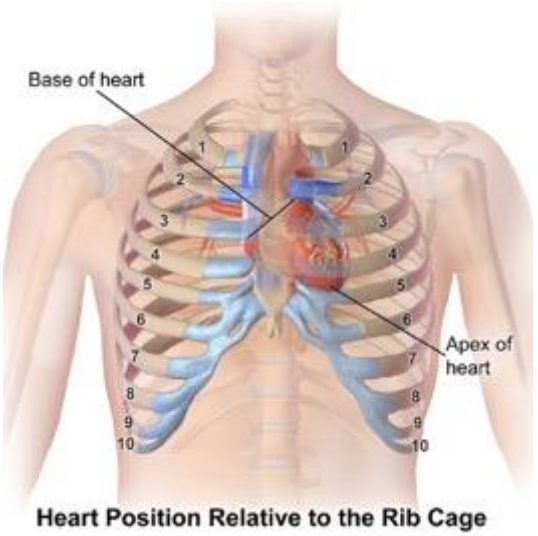


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Location & structure of the heart:

- Hollow muscular organ
- Approx the size of a fist
- Thoracic cavity – midline between sternum & vertebrae
- Sits at an angle under the sternum
- Broad ‘base’ at top – angled up and right
- Pointed ‘apex’ at the bottom – angled down and left
- CPR



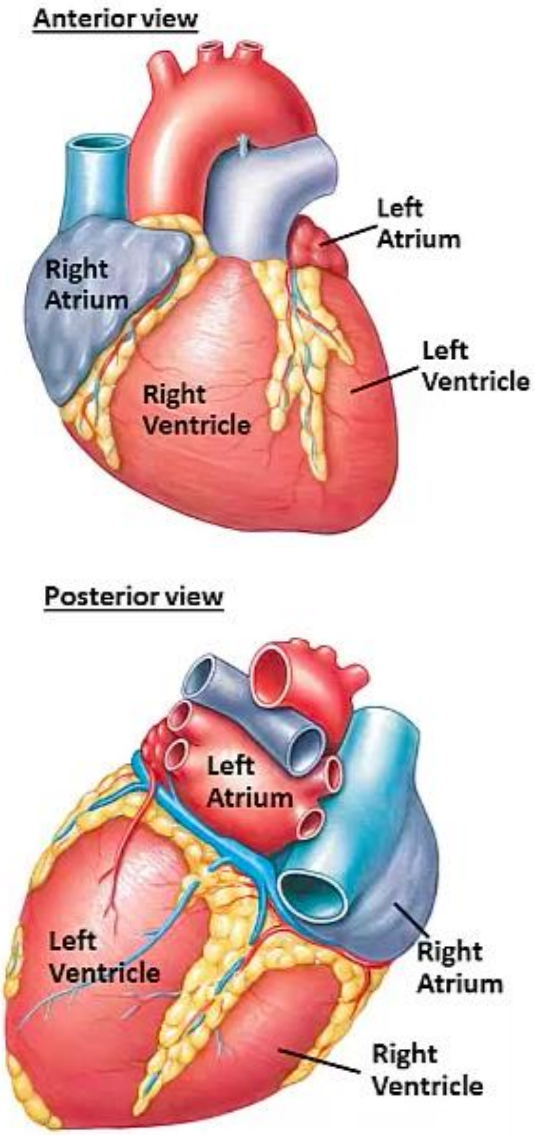


Heart Position Relative to the Rib Cage

Surface Anatomy:

Five surfaces:

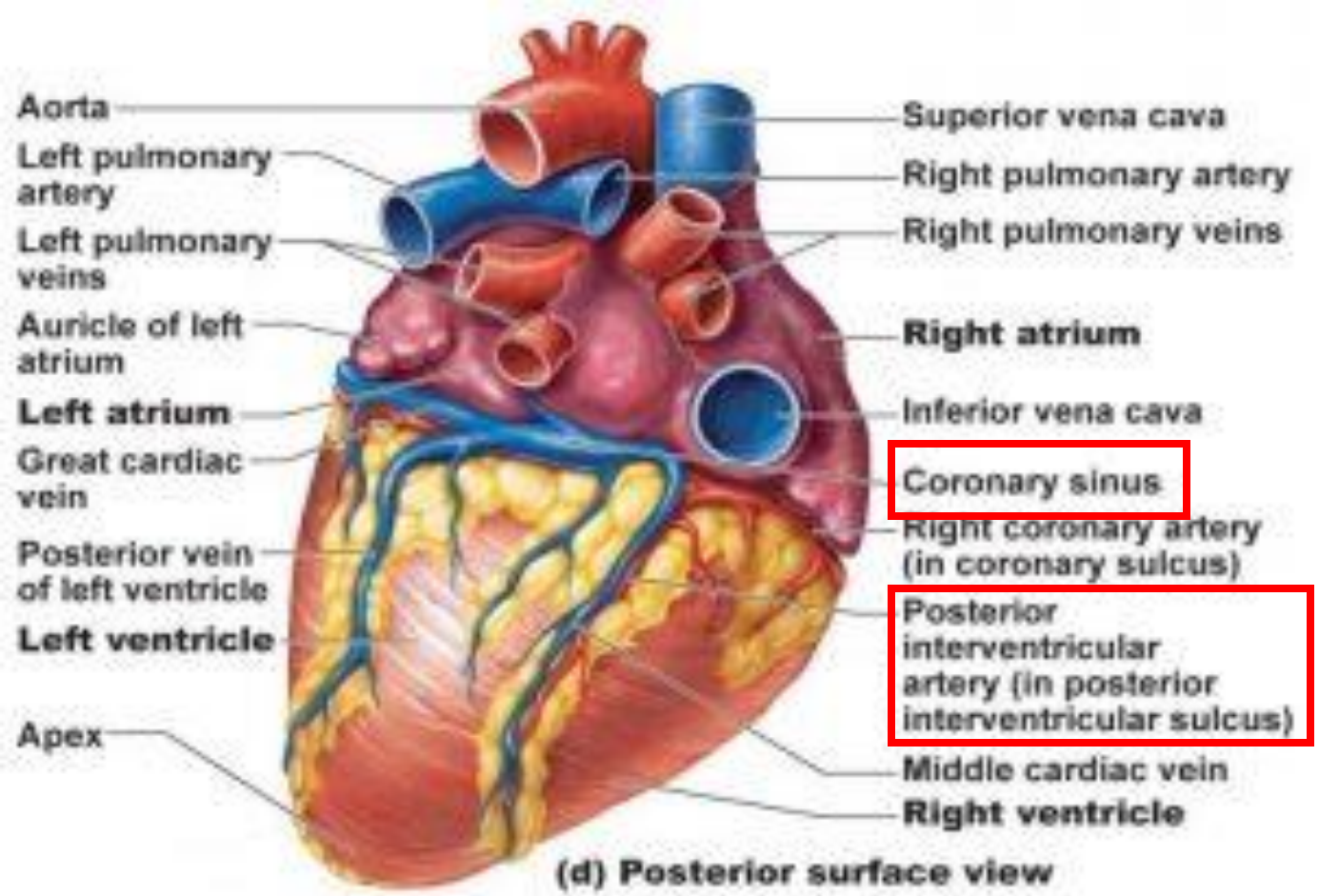
- Anterior (right ventricle)
- Posterior (left atrium)
- Inferior
- Right pulmonary
- Left pulmonary



Sulci of the Heart

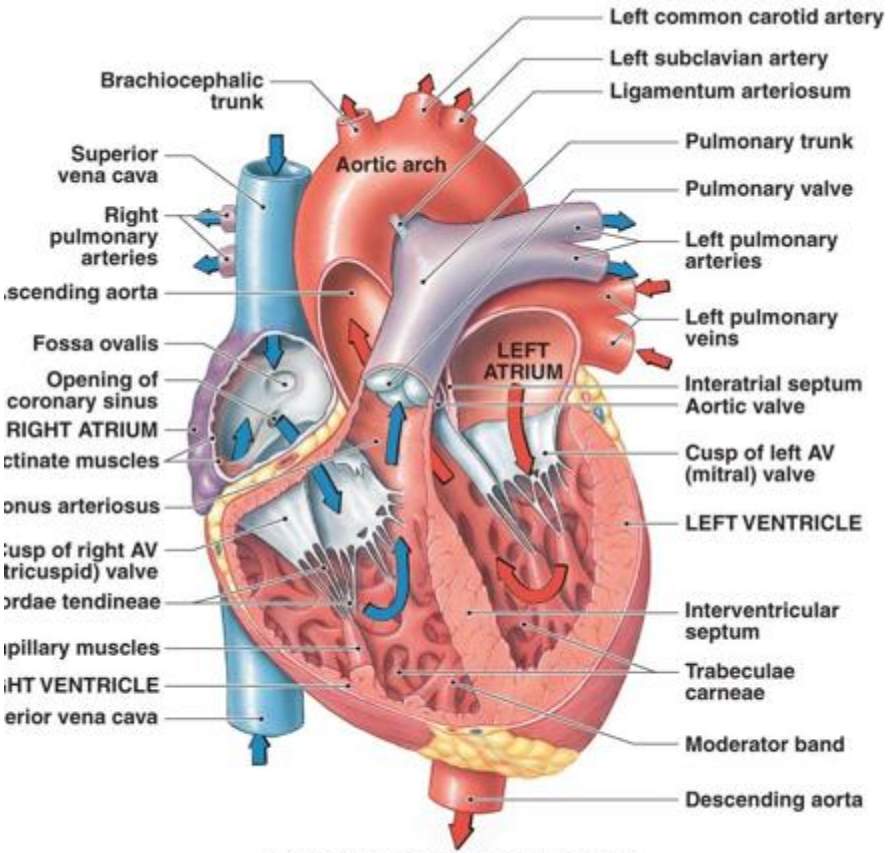
- Heart interior is divided into four chambers demarcated by grooves on the external surface (Sulci [plural]).
- **Coronary / atrioventricular sulcus** is a transverse band around the heart separating atria & ventricles (contains the right coronary artery).
- **Anterior** and **Posterior interventricular** sulci, vertically aligned bands on front and rear, separating the ventricles.

Sulci of the Heart



Structure of the heart continued:

- **4 chambers**
  - 2 upper 'collecting' chambers = ATRIA
  - 2 lower 'dispatching' chambers = VENTRICLES
- **Heart wall** (see next slide)
- **Muscular wall/septum** – separates right & left side of heart
  - Interatrial
  - Interventricular
- **Atrioventricular valves**
- **Pulmonary & Aortic Valves**



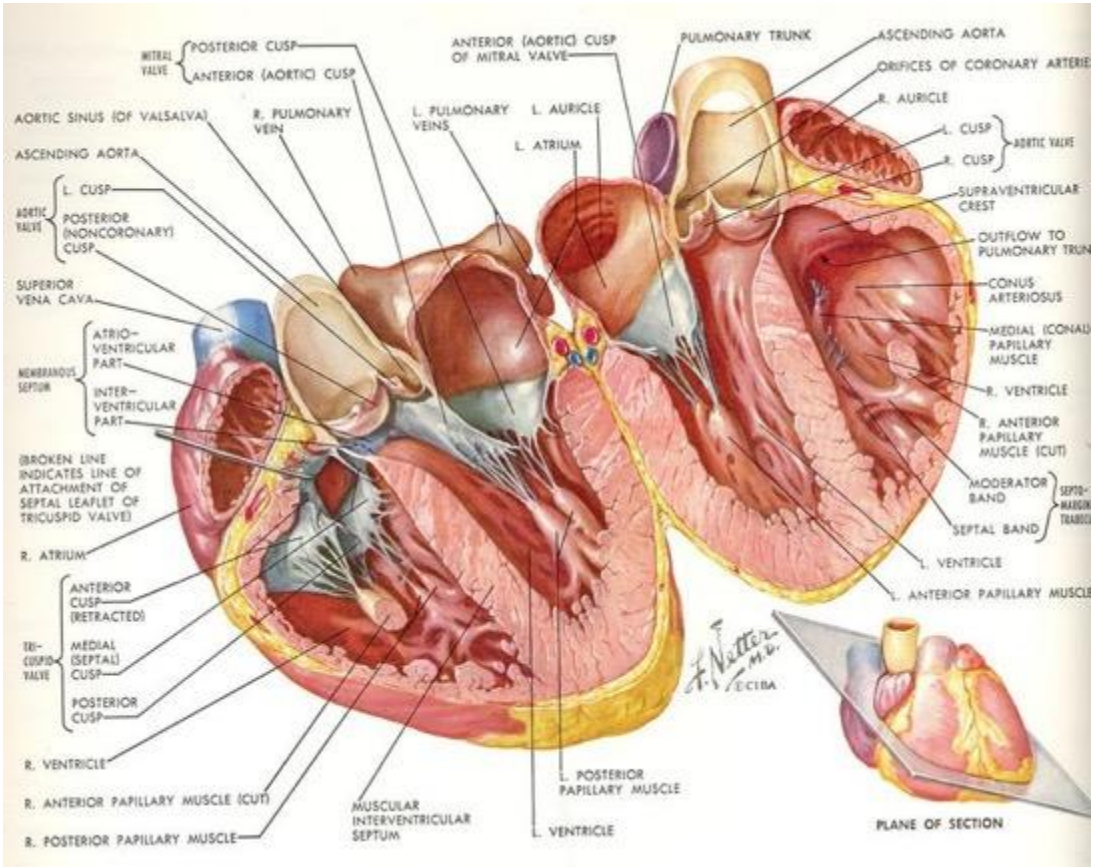
(a) Frontal section through the heart

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Chambers of the heart:

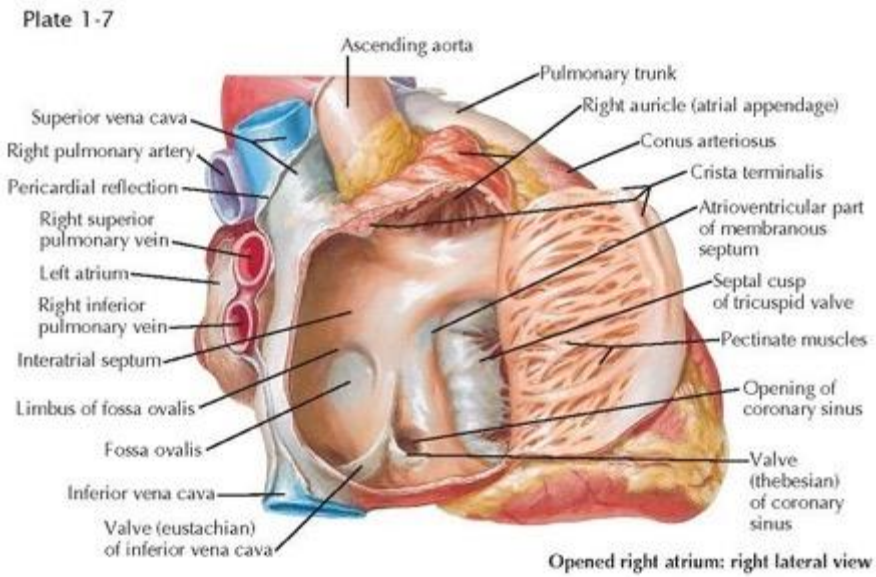
'Each chamber has intrinsic features that allows it to be described as 'morphologically right' or 'morphologically left' irrespective of location or distortion by malformation'

Gatzoulis et al, 2003



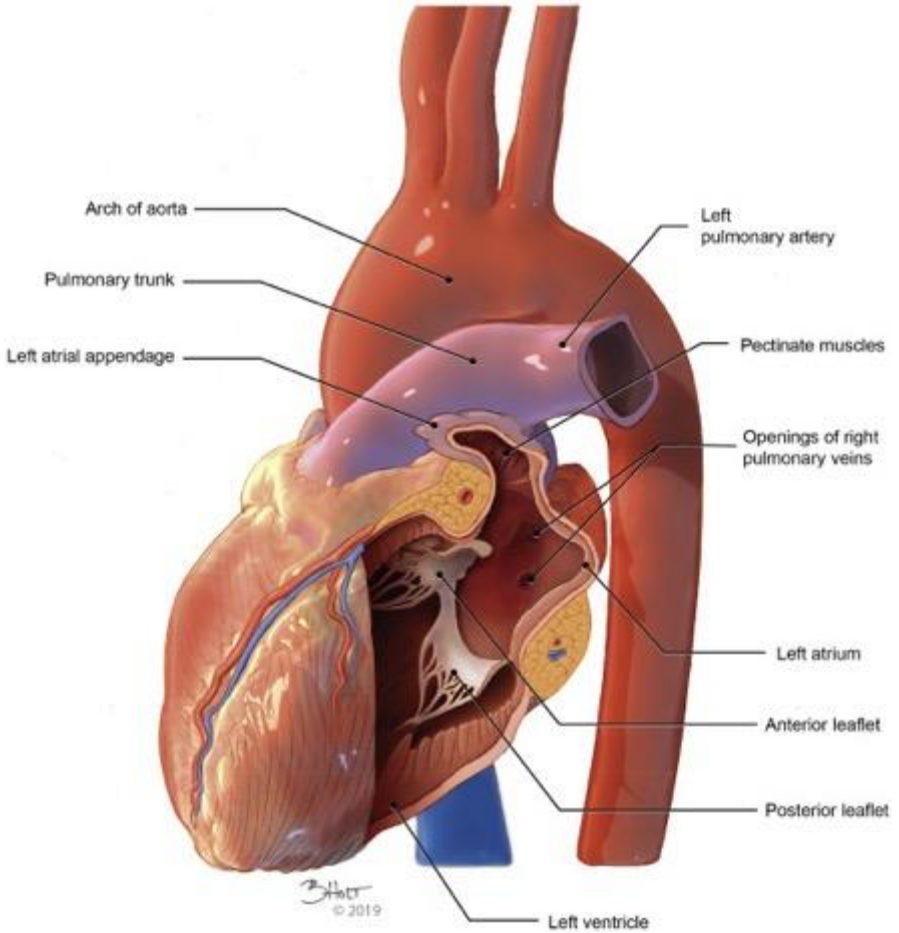
Right Atrium:

- Thin muscle wall
- Deoxygenated blood from SVC/IVC and coronary sinus (drains blood from myocardium)
- Most blood flow in occurs during inspiration (when RAP drops below SVC/IVC) (high-low)
- No valves in SVC/IVC therefore increased RAP results in systemic congestion



Left Atrium:

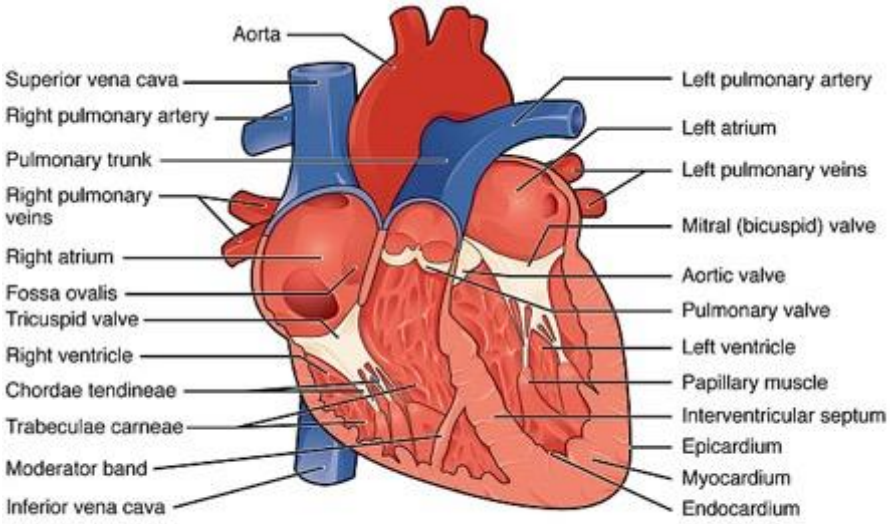
- Receives oxygenated blood from the lungs through pulmonary veins (PVs)
- Wall of LA is slightly thicker than RA and breathing does not affect its filling.



Ventricular chambers:

Three components:

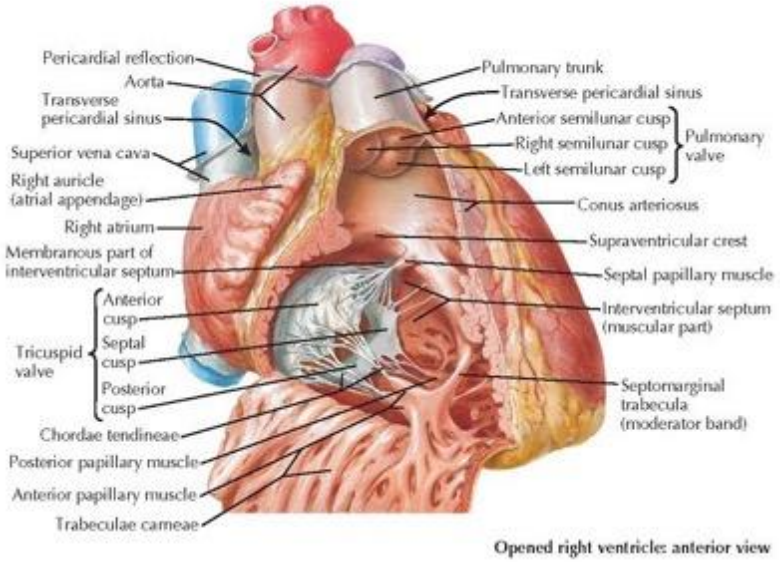
- Inlet
  - AV valve
- Outlet
  - Leads towards great arteries.
- Trabecular
  - Extends beyond the papillary muscles to the apex
  - Distinguishing marker



Anterior view

Right Ventricle:

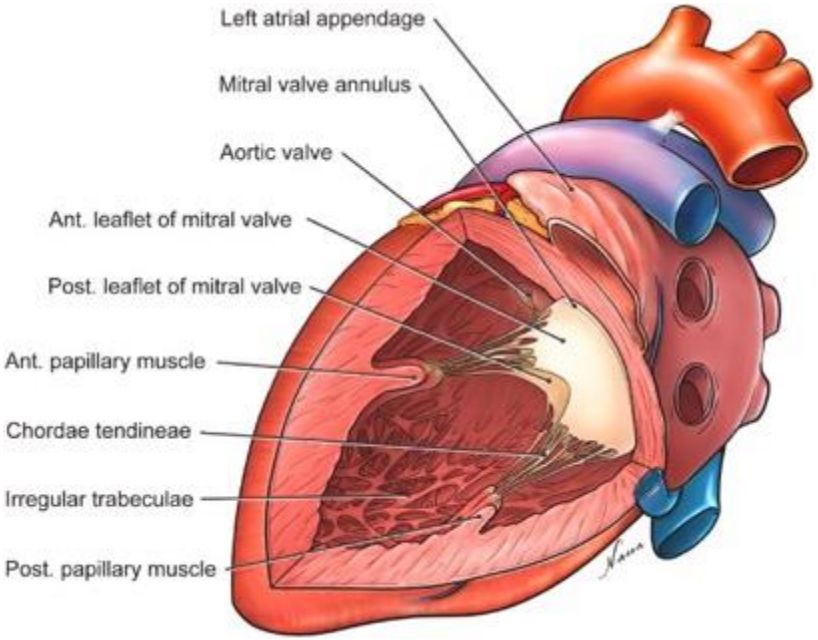
- Body of the RV
  - Inflow region consisting of - tricuspid valve, the chordae tendineae, the papillary muscle, and a heavily trabeculated myocardium
- Infundibulum
  - a smooth outflow region
- ‘Separated’ by four muscular bands
- Little pressure is needed to pump blood to the lungs.
- The RV generates less than ¼ stroke work of the left ventricle.

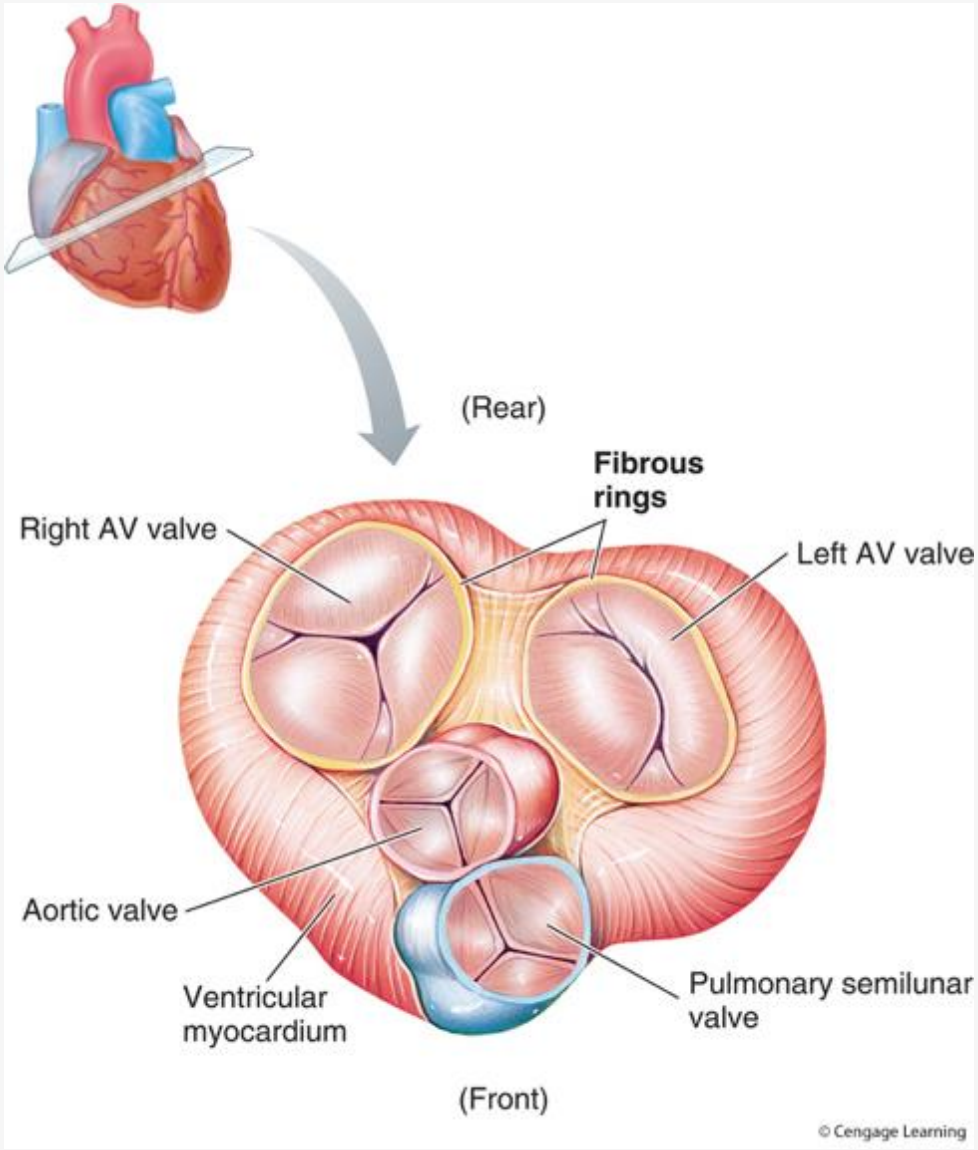


Opened right ventricle: anterior view

Left Ventricle:

- Thick muscular wall
- Receives blood from the LA through MV and ejects through AV to the systemic circulation via aorta.
- Pressure in LV is high.
- The ventricular septum - houses electrical conduction tissue and provides stability for the ventricles during contraction.



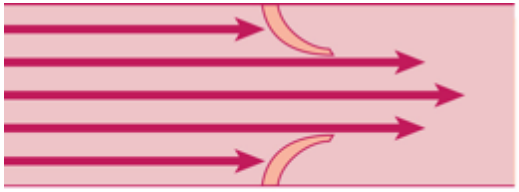


Fibrous skeleton:

- 4 interconnecting rings of dense connective tissue
- Surround and support valves – on same plane
- Separates atria & ventricles
- Rigid structure for cardiac muscle to attach to
- Twisted structure supports more efficient pumping

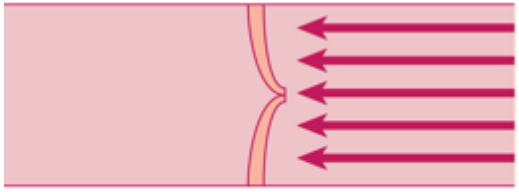
Valves:

- Blood flows through the heart in one fixed direction
- 4 x 1 way valves ensure no backwash
- Valves open and close passively due to pressure differences – ‘one way door’



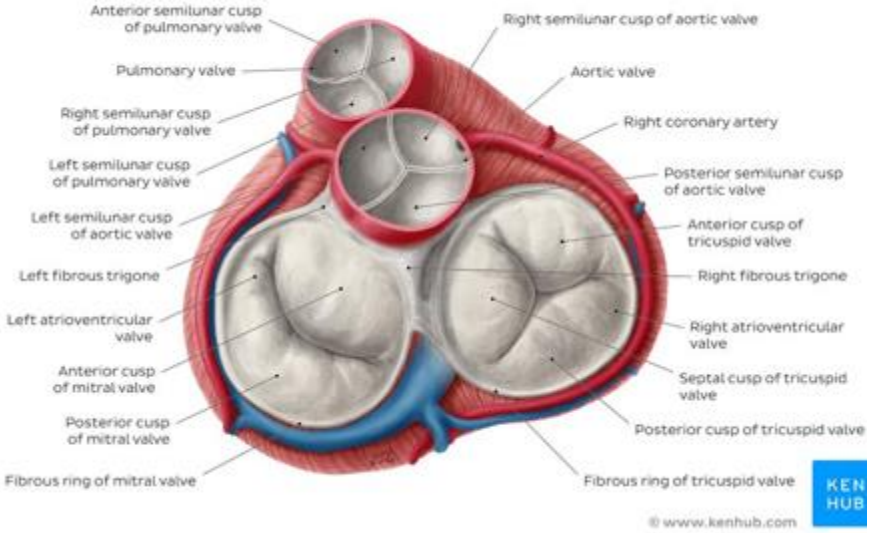
Valve opened

When pressure is greater behind the valve, it opens.



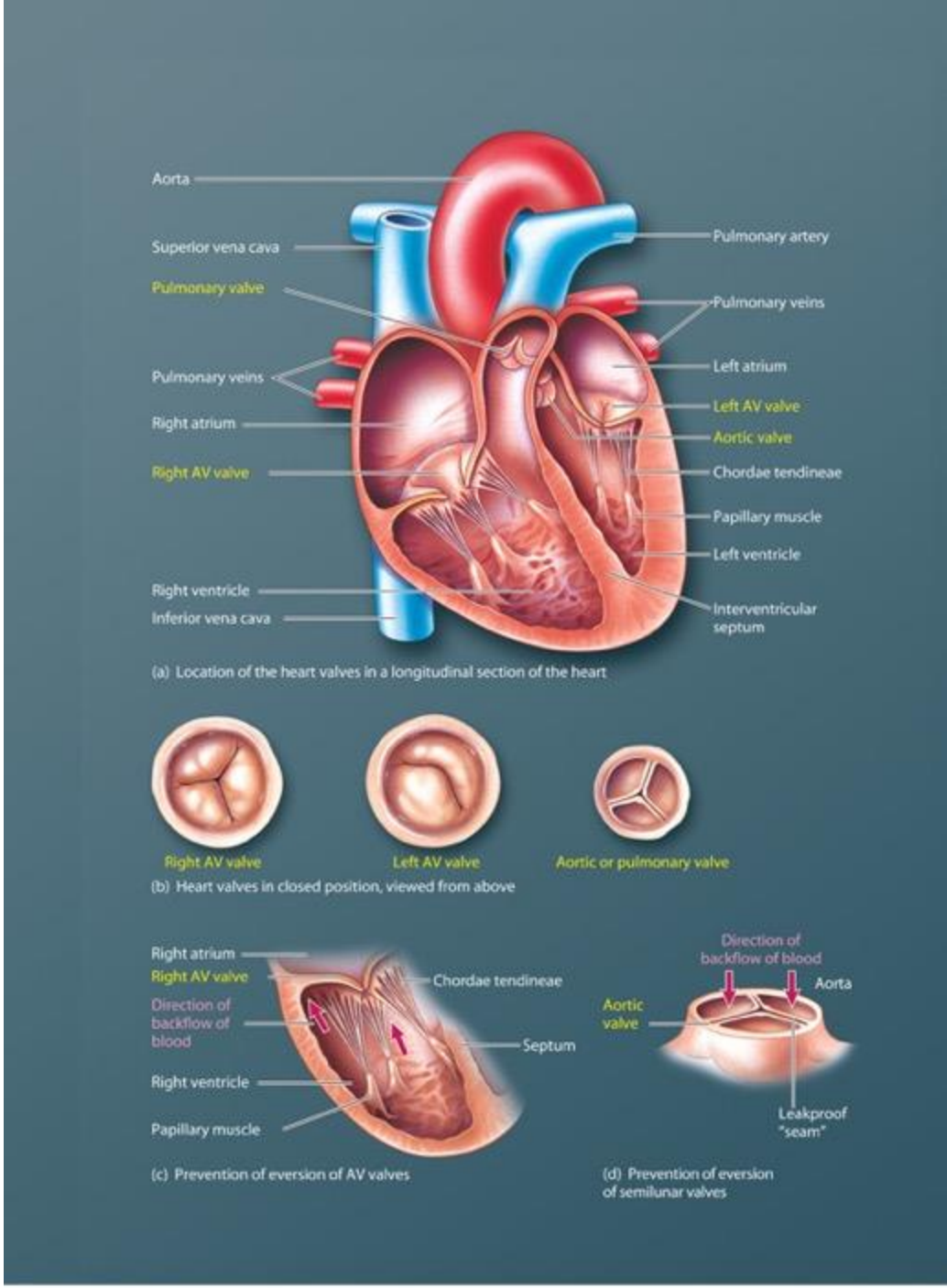
Valve closed; does not open in opposite direction

When pressure is greater in front of the valve, it closes. Note that when pressure is greater in front of the valve, it does not open in the opposite direction; that is, it is a one-way valve.



Atrioventricular valves:

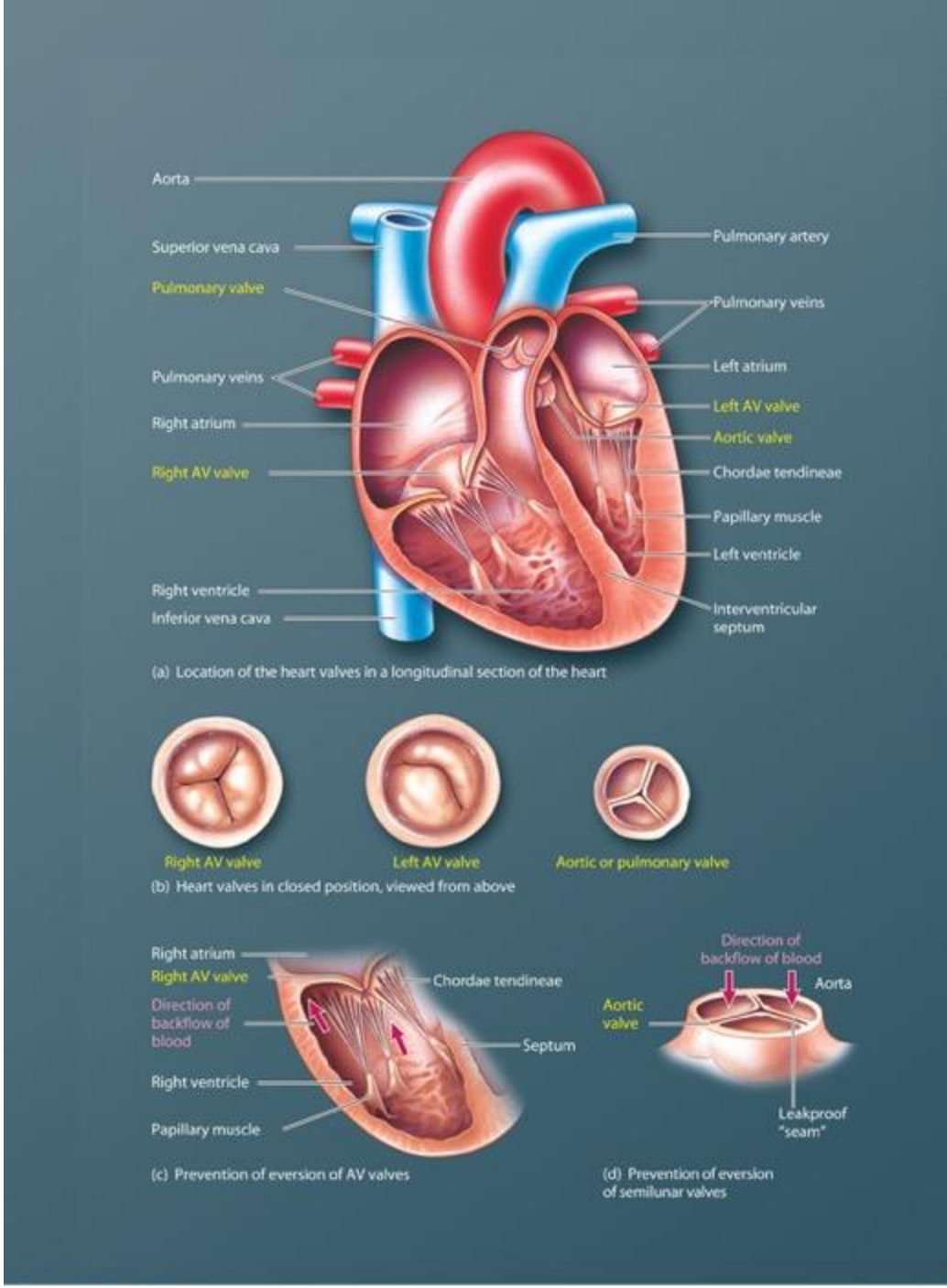
- Between atria & ventricles
- Atrial pressure > ventricle pressure
  - Ventricular filling
  - Prevents backflow during ventricular emptying
- Right AV valve = tricuspid
- Left AV valve = bicuspid/mitral



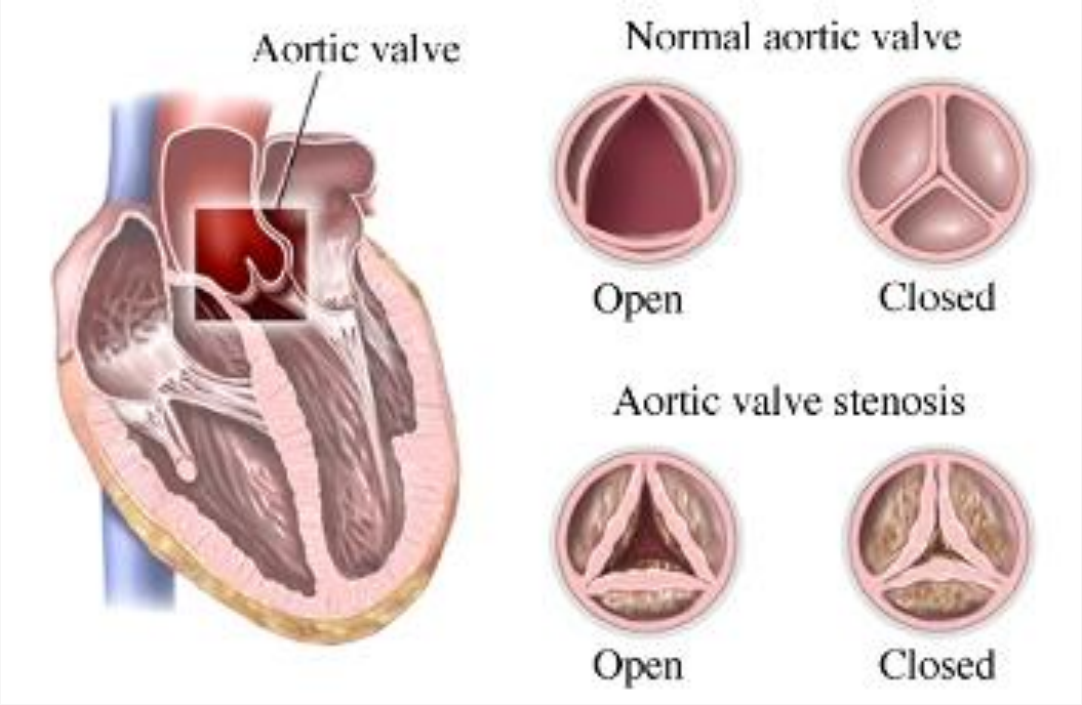
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Semilunar Valves:

- Between ventricles and major arteries
- Semilunar – 3 cusps, like half moons
- Forced open when ventricular pressure > pressure in aorta/pulmonary artery
  - Ventricular contraction & emptying
- Close when ventricles relax

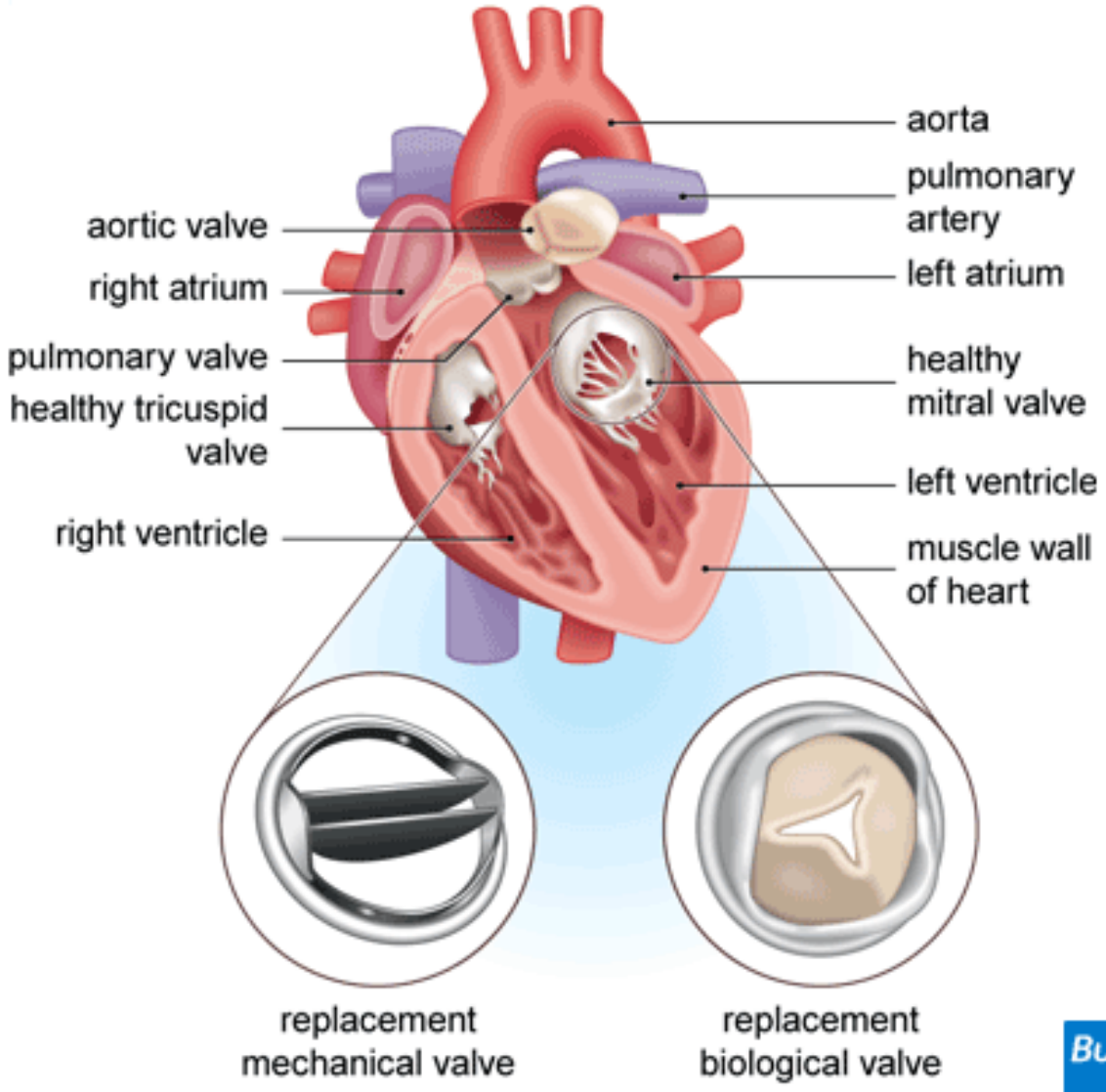


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Pathology link – Valve disease:

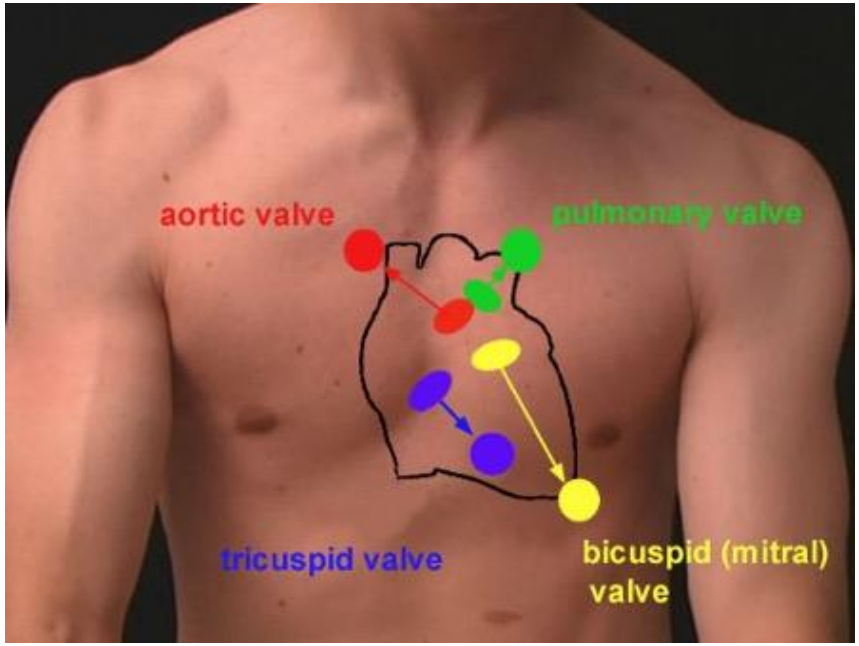
- **Regurgitation**: blood flows backwards through leaky valve leads to progressive overload and then heart failure
- **Stenosis**: narrowing of the valve, caused by scarring, calcification or congenital.
- **Causes:** congenital heart disease**,** rheumatic fever, cardiomyopathy, damage to the heart muscle from an MI, getting older, a previous infection with endocarditis
- **Treatment** : TAVI (**transcatheter aortic valve implantation**), valve repair or replacement.



The types of heart valve replacement

Heart sounds

- Sounds of the heart valves closing - commonly termed ‘lub dub’.
- ‘lub’ sound of the AV valves closing
- ‘dub’ sound of closing of the semilunar valves

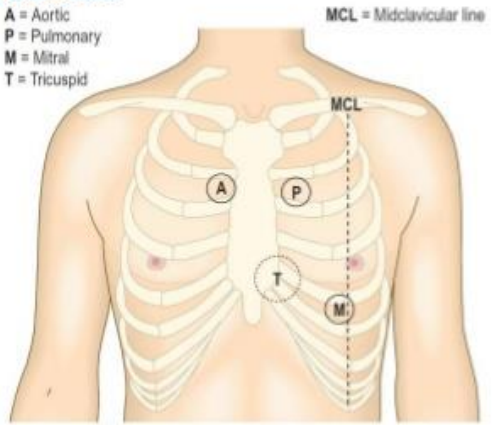


Auscultation

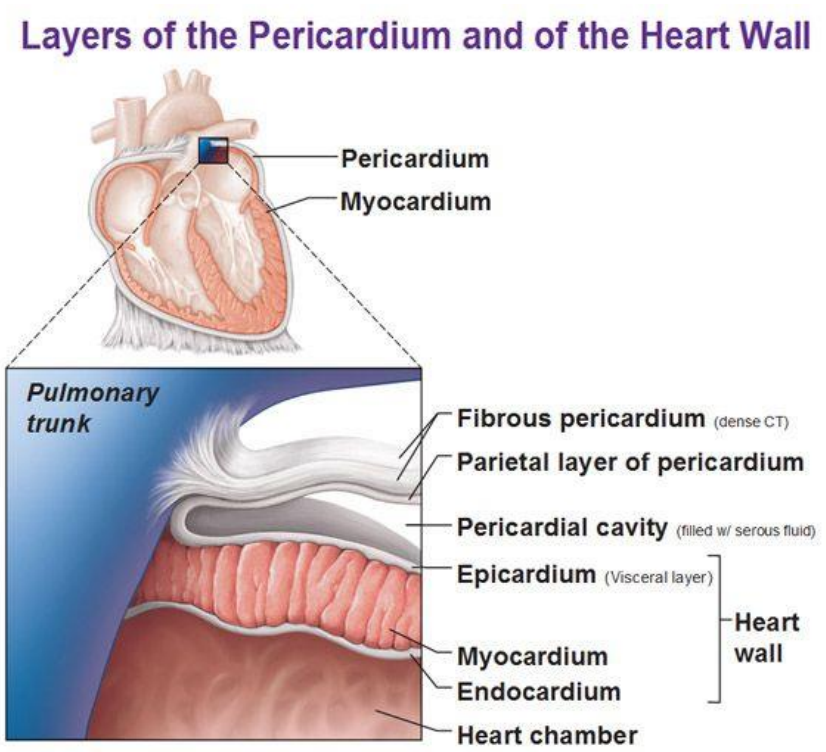
✓ Auscultation for Heart Sounds

First sound (S1)
This corresponds to mitral and tricuspid valve closure at the onset of systole.

Second sound (S2)
This corresponds to aortic and pulmonary valve closure following ventricular ejection.



Heart consists of 4 layers

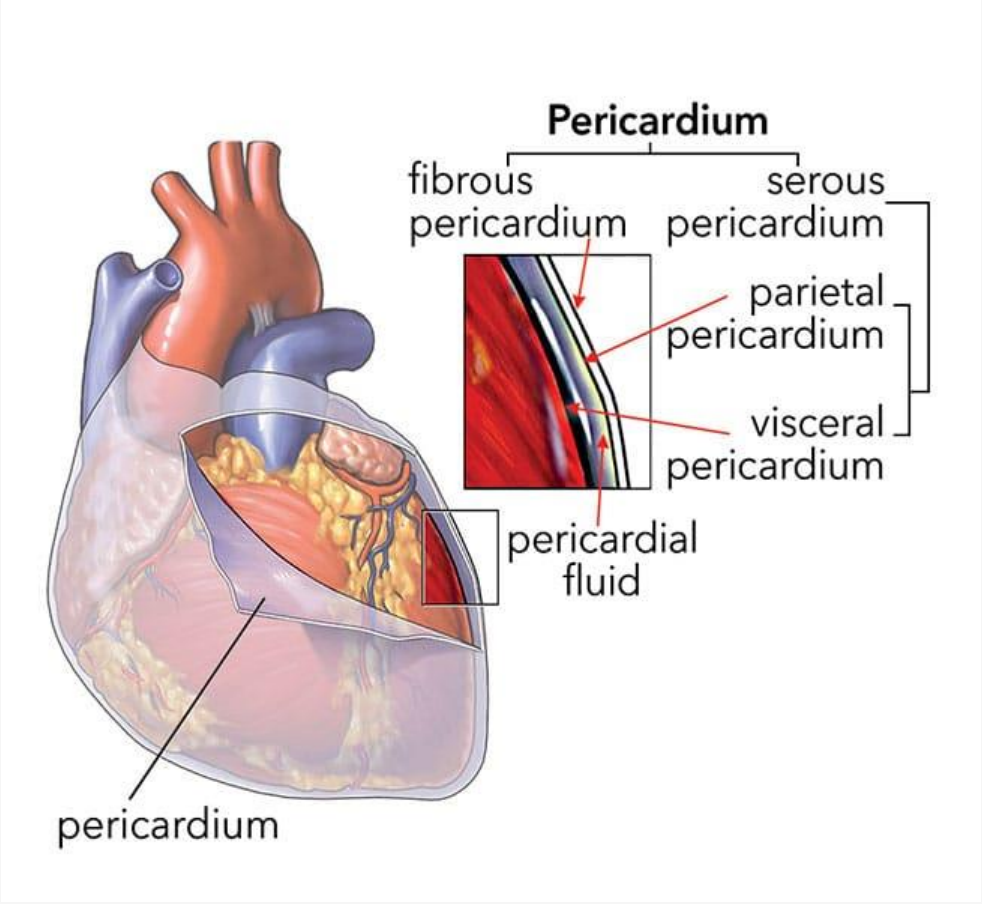


3 layers of the heart wall:

- **Inner layer = endocardium/endothelium**
  - Thin internal layer
  - Epithelial tissue
  - Lines entire circulatory system
- **Middle layer = myocardium**
  - Cardiac muscle
  - Bulk of heart wall
- **Outer layer = epicardium**
  - Thin external layer

Pericardial Sac

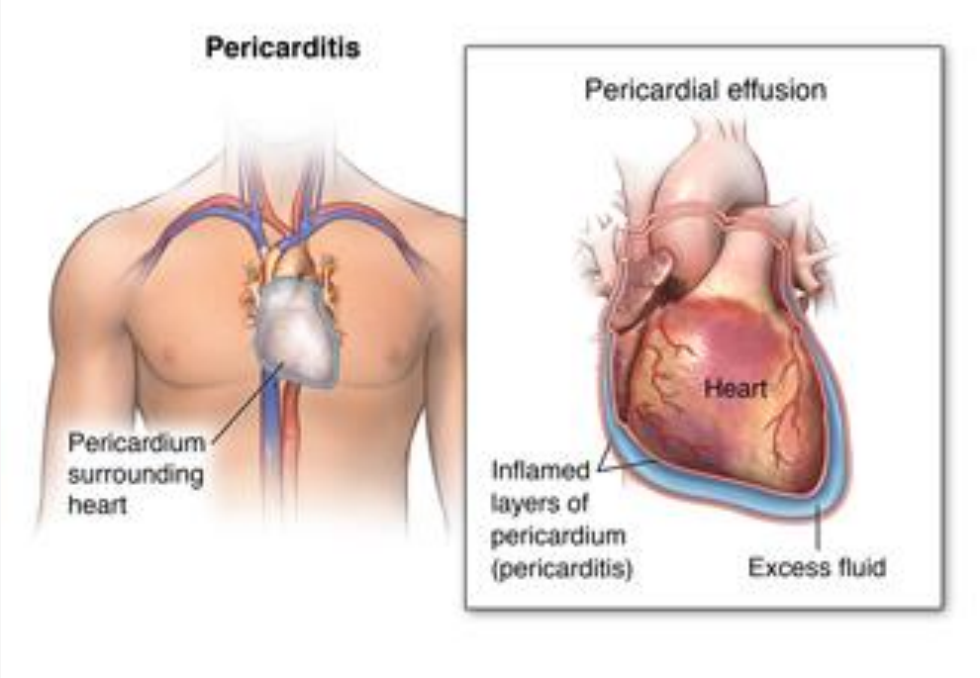
- Double walled, membranous outer sac



Pericardium:

The pericardium has several protective functions:

- Is fused to the diaphragm so downward movement during inspiration pulls heart more vertically
- Keeps the heart contained within the chest cavity,
- Prevents the heart from over-expanding when blood volume increases,
- Limits heart motion,
- Reduces friction between the heart and surrounding tissues, and.
- Protects the heart against infection.

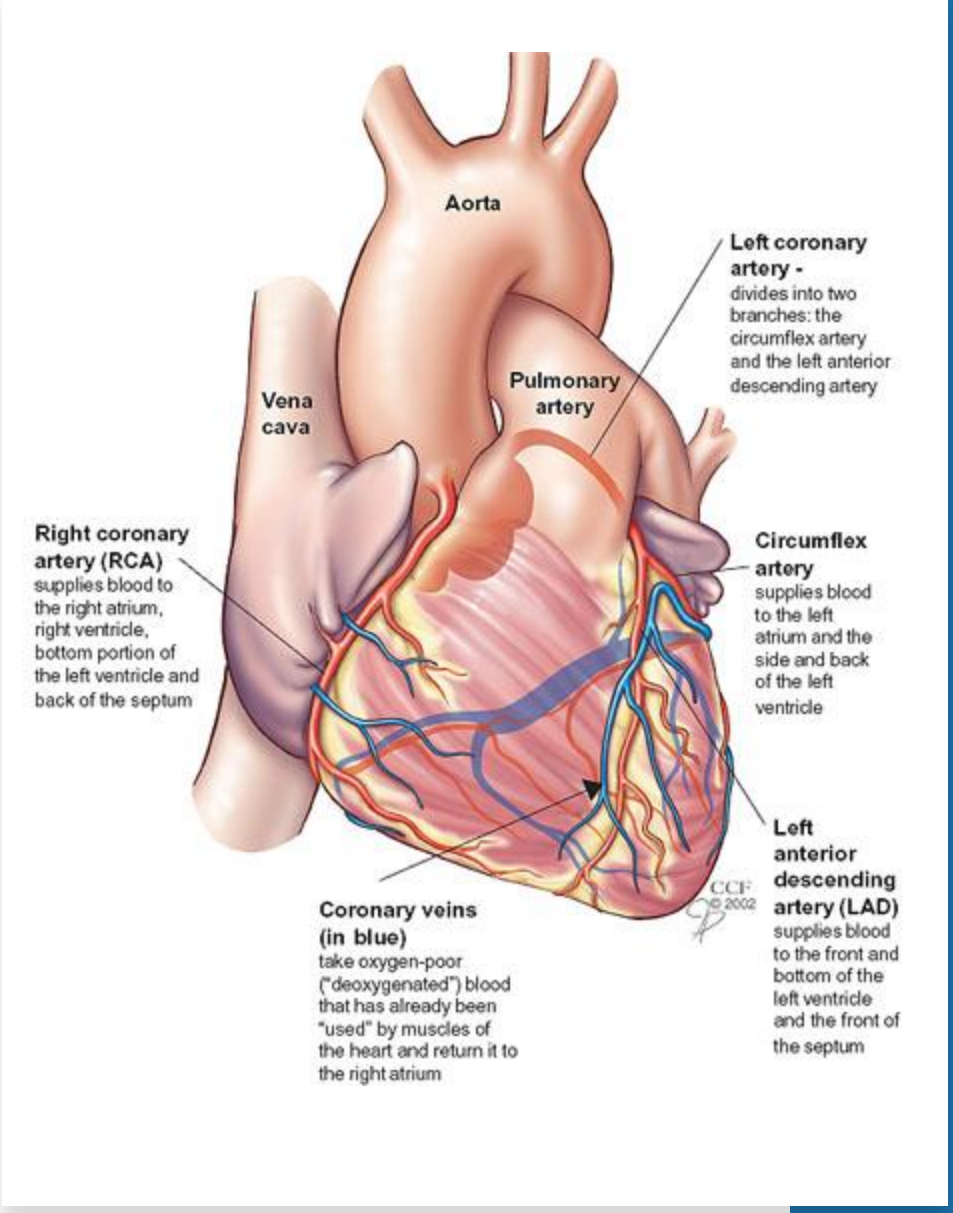


Pathology link – Pericarditis:

- Inflammation of the pericardial sac
- Potential leak of blood or fluid between layers
- Painful friction
- Often due to viral/bacterial infection

Blood supply of the heart – Coronary Arteries

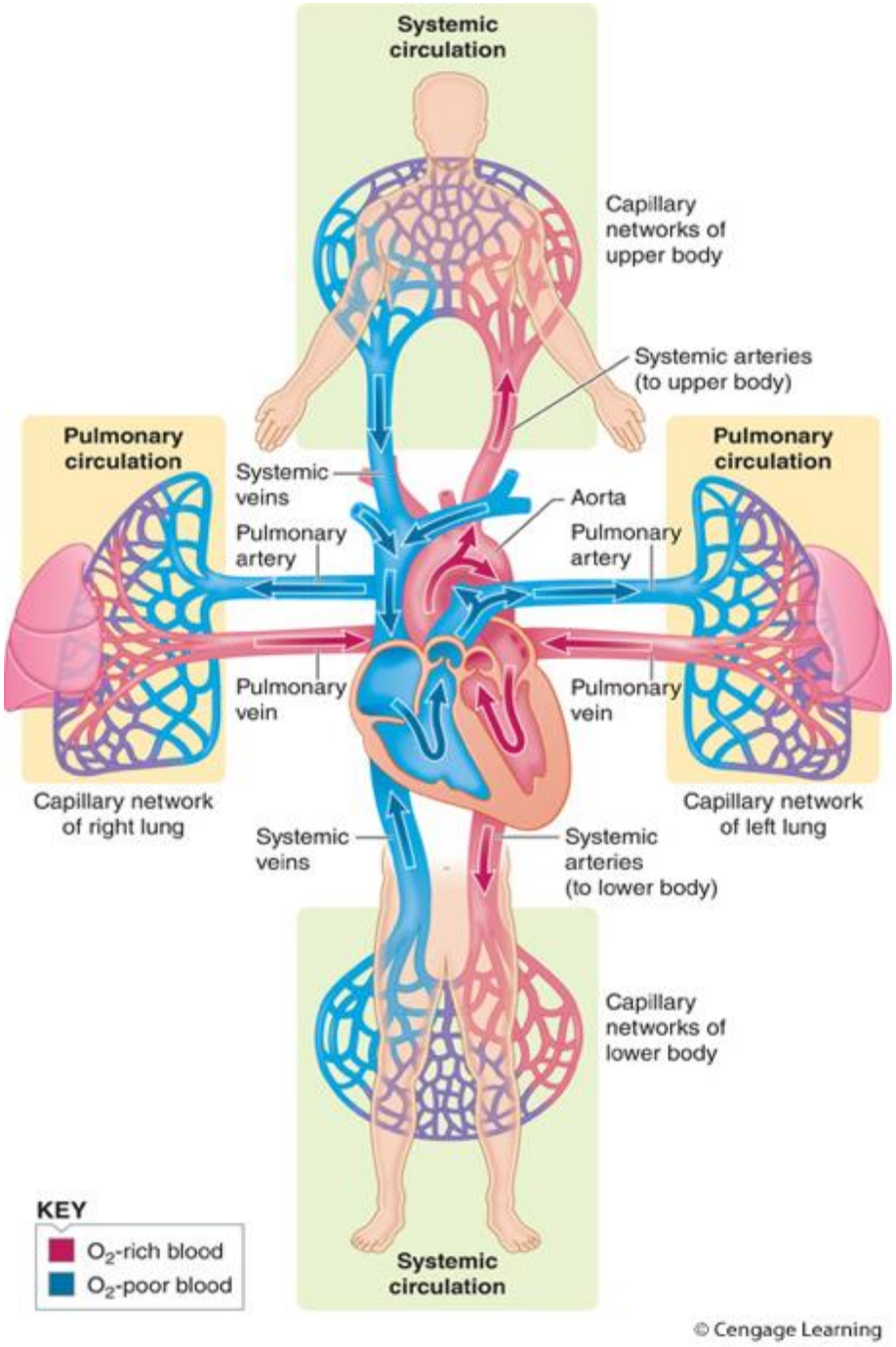
- Heart has its own blood supply from the Coronary arteries (crown the apex)
- Coronary arteries wrap around outside of heart with small branches penetrating into heart muscle for blood
- 2 major coronary arteries branch off from the aorta near where aorta and left ventricle meet (**Left and Right Main coronary arteries**)
- LMCA supplies left side of heart (LA and LV) and has **L anterior descending artery** (front of left side of heart) and **circumflex artery** (outer side and back of heart)
- R Coronary artery supplies blood to the right ventricle, the right atrium, and the SA (sinoatrial) and AV (atrioventricular) nodes. Also supplies middle (septum)
- Coronary veins take blood that has been used by the muscles of the heart and return it to Right atrium



Summary of Circulation:

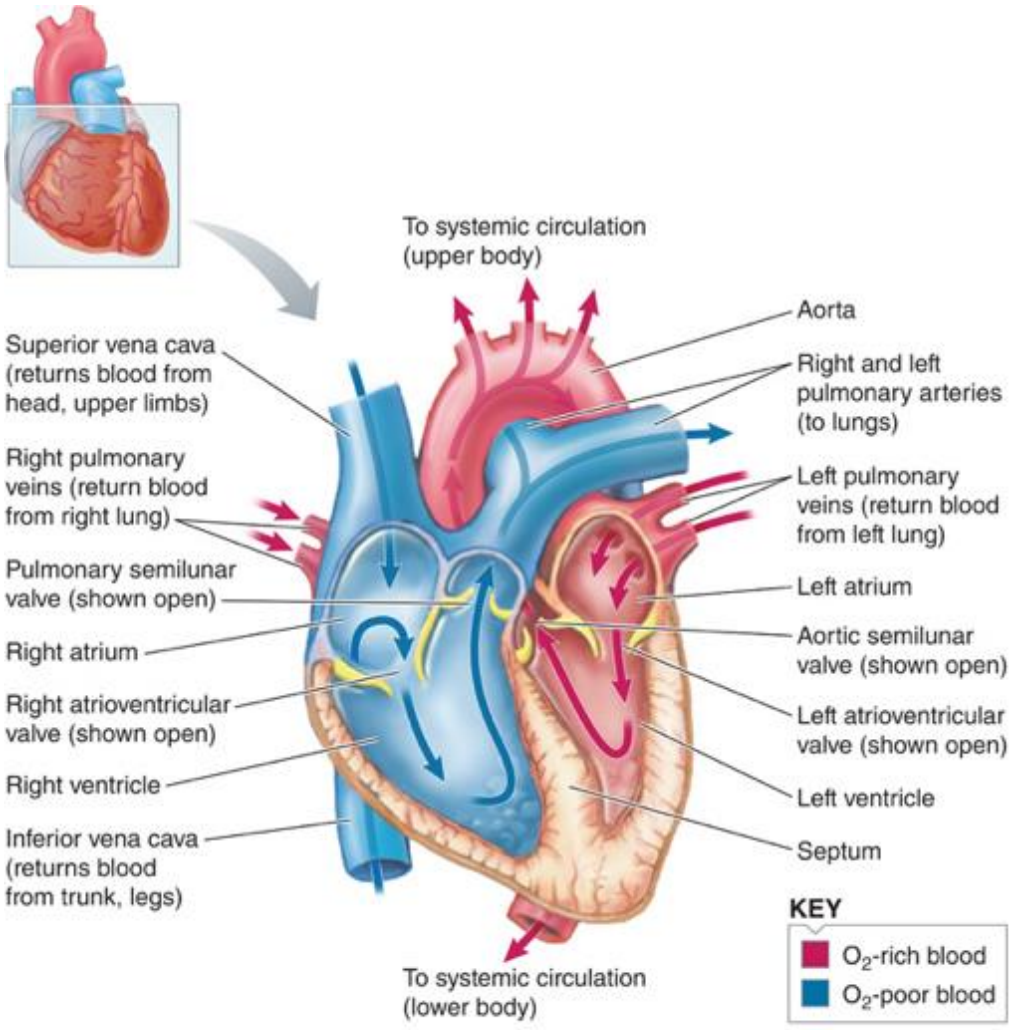
2 separate vascular loops:

- **Pulmonary Circulation**
  - Closed loop
  - Carries blood between heart & lungs
- **Systemic Circulation**
  - Carries blood between heart and all body systems



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Flow of blood through the heart:

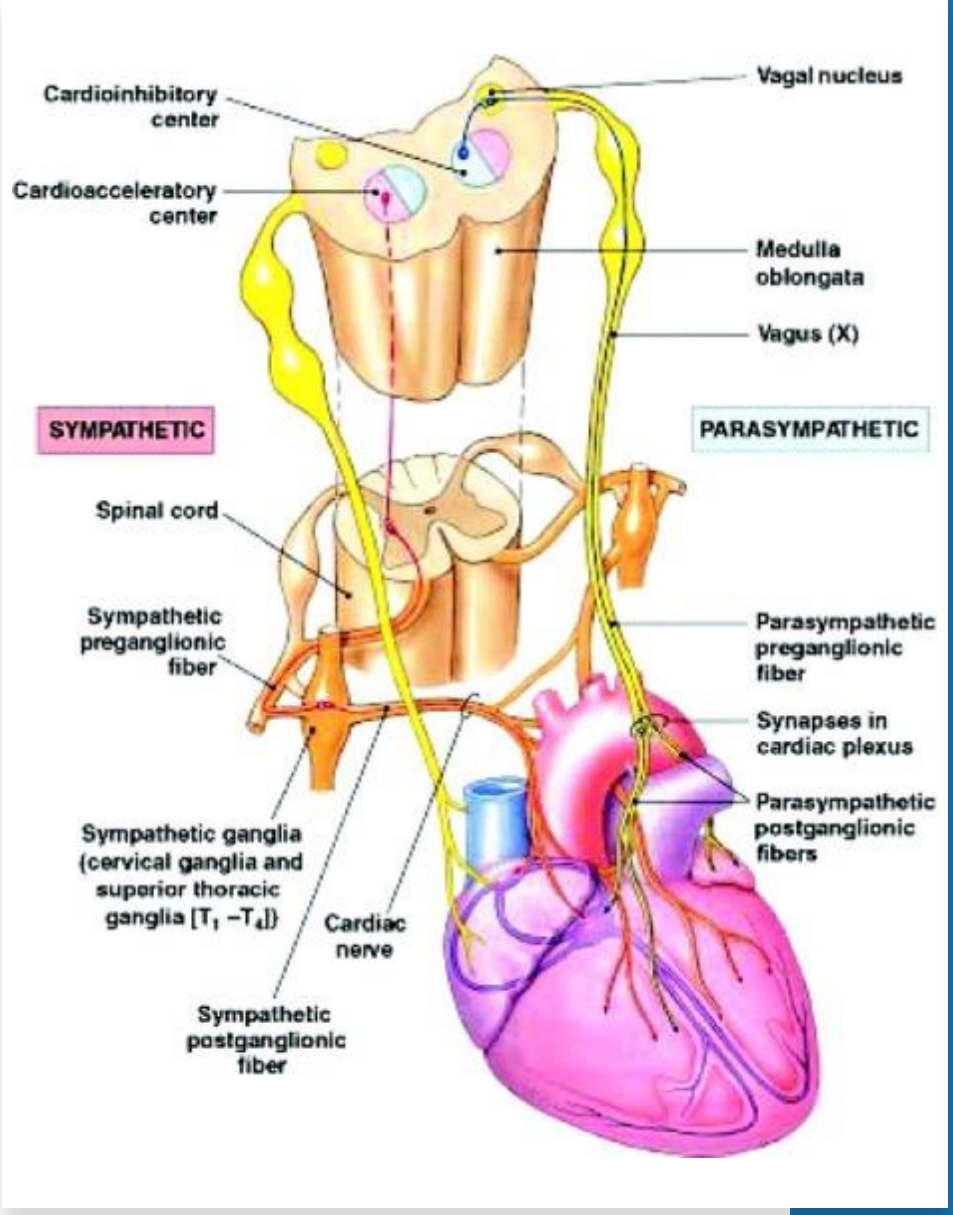


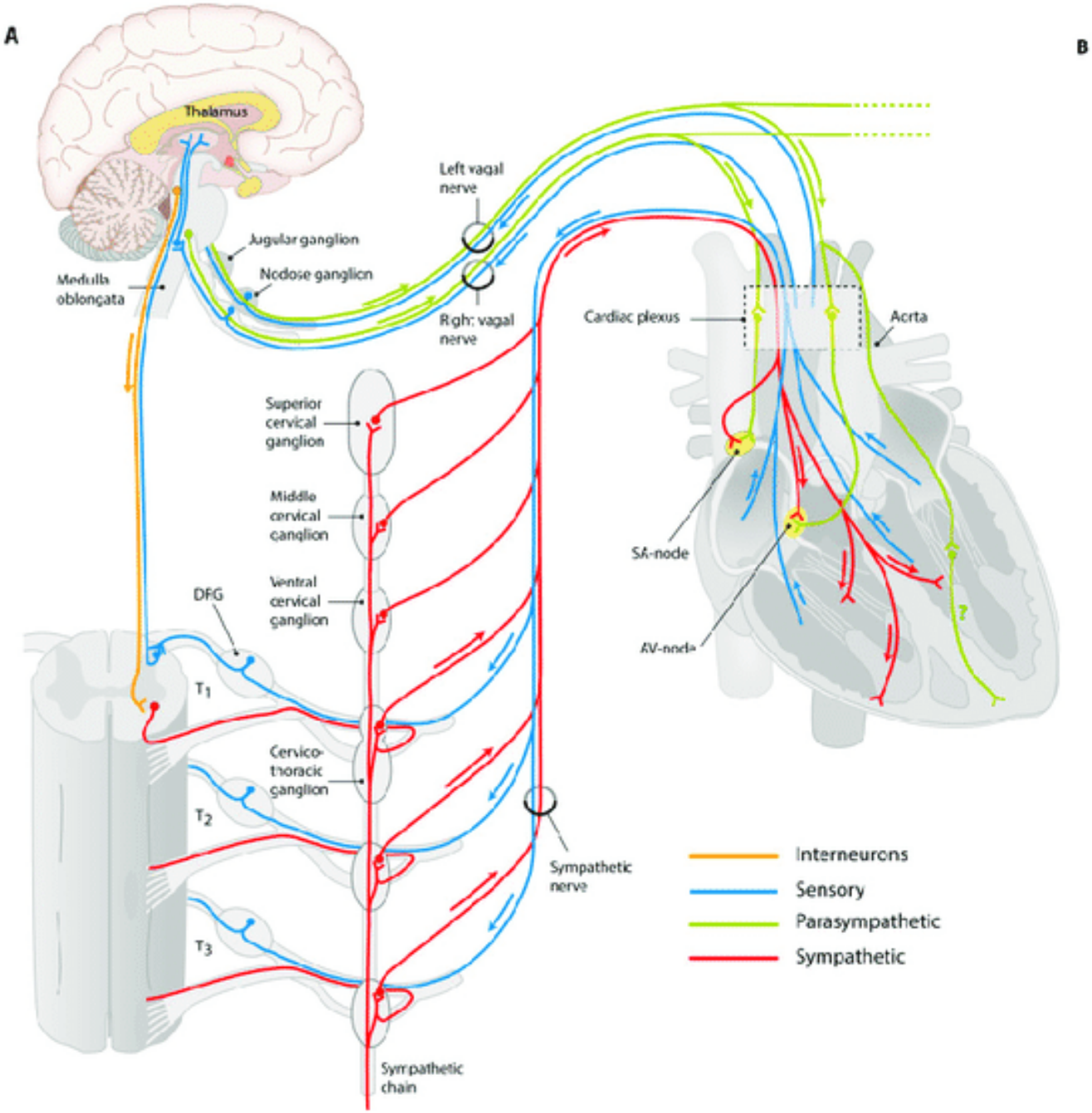
(a) Blood flow through the heart

- Deoxygenated blood from body > Right atrium via superior & inferior vena cava
- Right atrium > Right ventricle (via tricuspid valve)
- Right ventricle > Pulmonary artery (via pulmonary valve)
- Pulmonary artery > L & R branches > Lungs
- Oxygenated blood from lungs > Left atrium via R & L branches of pulmonary veins
- Left atrium > Left ventricle (via mitral valve)
- Left ventricle > Aorta (via aortic valve)
- Aorta > Rest of body

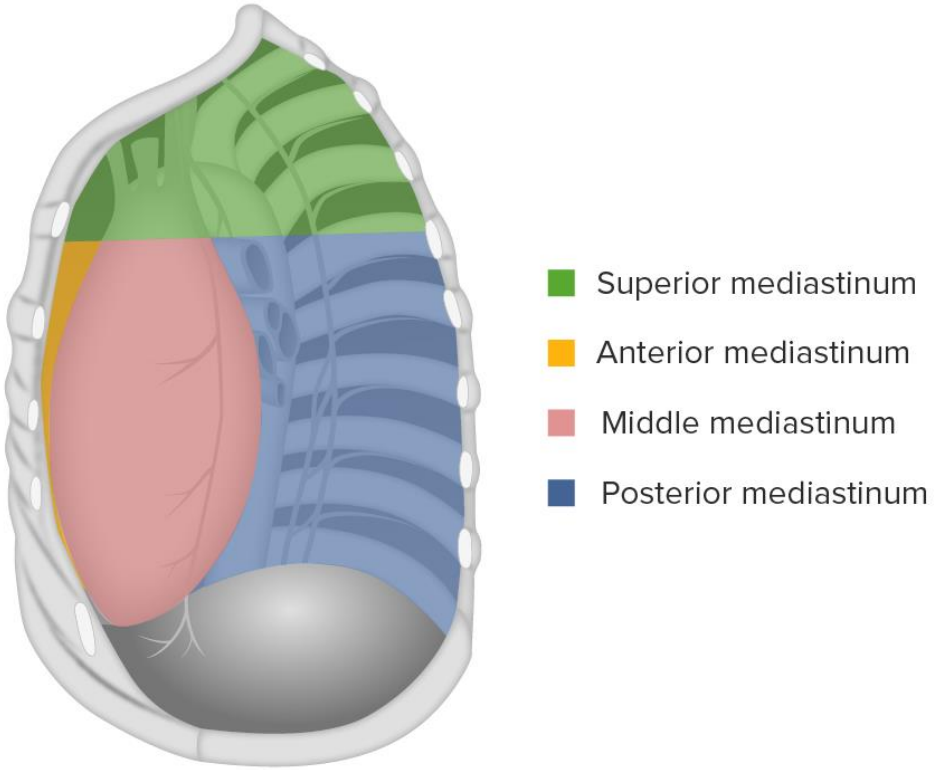
Nerve supply of the heart:

- Located in the MEDULLA (brain stem)
- Efferent Pathway = Autonomic nervous system
- Cardiovascular control centre alters the ratio of sympathetic : parasympathetic activity on the effector organs
- Effector organs = heart & blood vessels



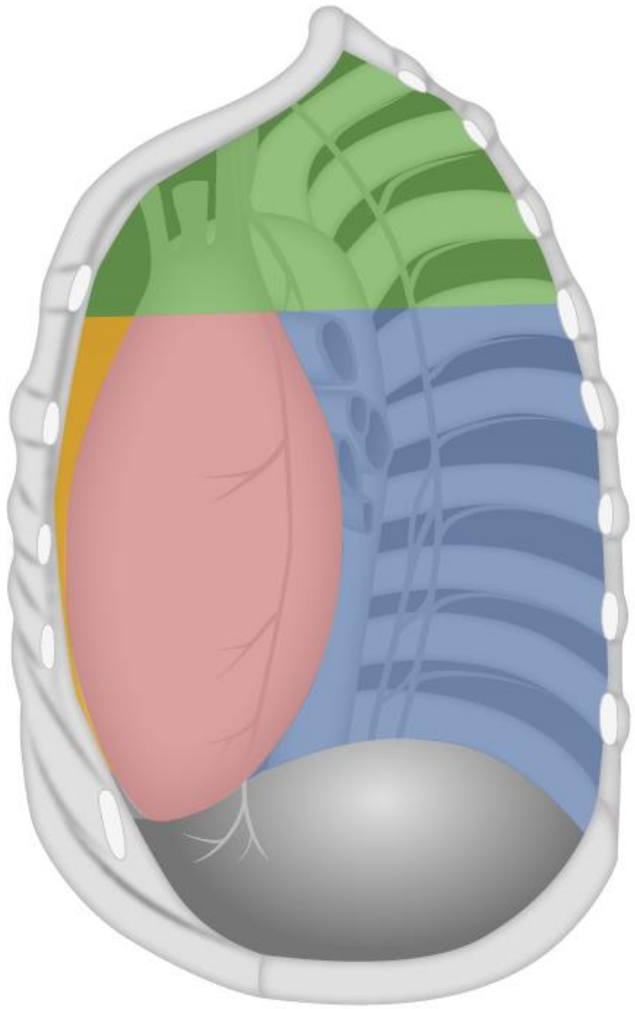

A
B
Thalamus
Left vagal nerve
Jugular ganglion
Nodose ganglion
Medulla oblongata
Right vagal nerve
Cardiac plexus
Aorta
Superior cervical ganglion
Middle cervical ganglion
Ventral cervical ganglion
DRG
SA-node
AV-node
T1
T2
T3
Cervico-thoracic ganglion
Sympathetic nerve
Sympathetic chain
Interneurons
Sensory
Parasympathetic
Sympathetic

Mediastinum



- Chest space holding heart and several significant structures
- Located mid thoracic between left and right plural cavities

Mediastinum
- Borders, Organs & Nerves



Superior:

Bordered by the thoracic outlet superiorly

Transverse thoracic plane or sternal angle inferiorly

Medial border of the pleural sacs laterally

Dorsal surface of the sternum anteriorly

Ventral surface of the first four thoracic vertebral bodies posteriorly.

Organs: thymus, trachea, oesophagus

Nerves: left and right vagus, recurrent laryngeal, cardiac, left and right phrenic

Mediastinum - Borders, Organs & Nerves

Anterior:

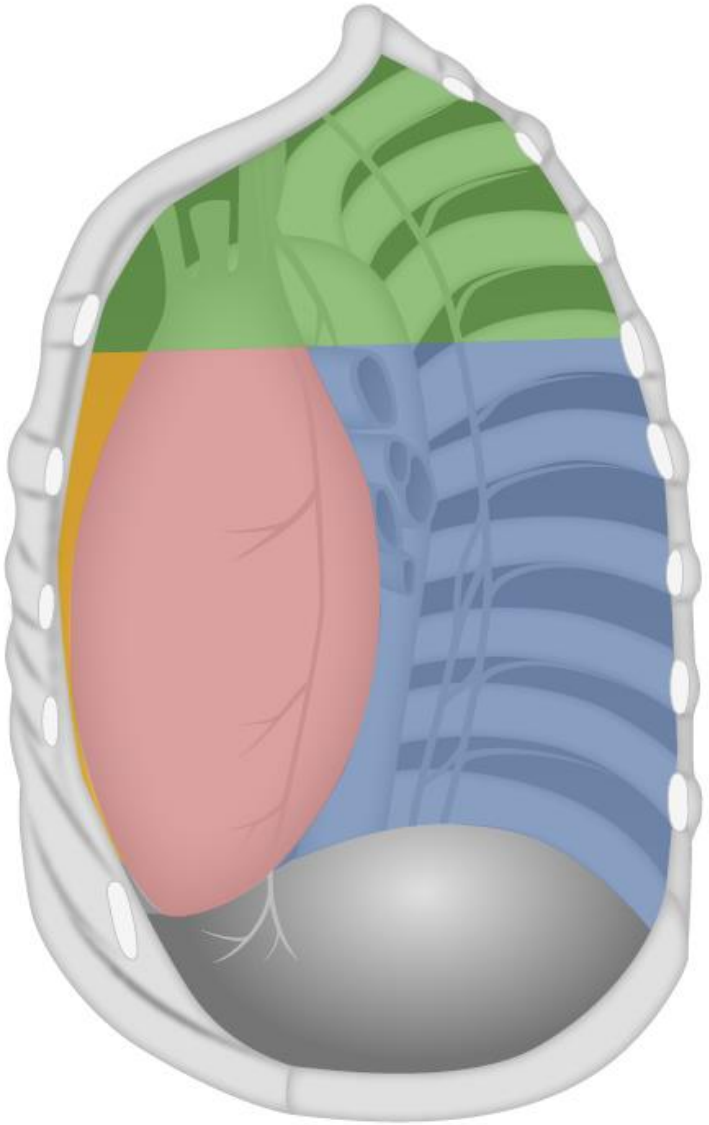
Bordered by the pericardium posteriorly

Medial border of the pleural sacs laterally

Sternum, transversus thoracis muscles, and 5th, 6th and 7th left costal cartilages anteriorly.

Organs: thymus

Nerves: none

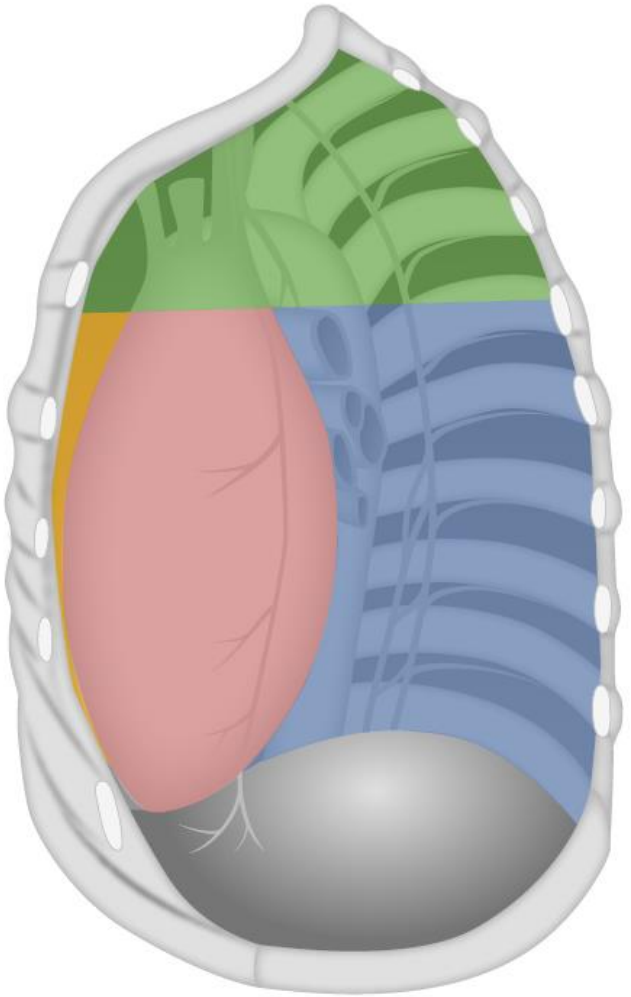


Mediastinum - Borders, Organs & Nerves

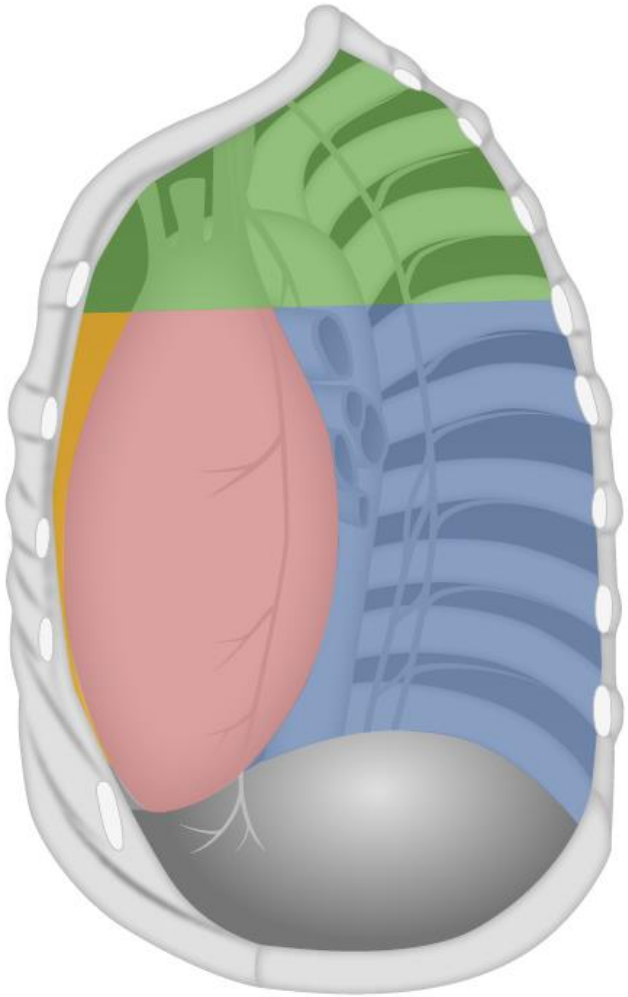
Middle:
Formed by the borders of the pericardial sac anteriorly and posteriorly
Reflected to the medial borders of the pleural sacs bilaterally
Transverse thoracic plane superiorly
Thoracic surface of the diaphragm inferiorly

Organs: the heart and its great vessel roots, trachea and main bronchi

Nerves: phrenic, vagus, sympathetics



Mediastinum - Borders, Organs & Nerves



Posterior:

Bordered by the pericardium anteriorly

Thoracic surface of the diaphragm inferiorly

Transverse thoracic plane superiorly

5th – 12th thoracic vertebral bodies posteriorly

Pleural sacs laterally

Organs: oesophagus

Nerves: vagus, splanchnic, sympathetic chain

Thank you for listening.

Please submit any questions via Weblearn.